

Experiment 4

Programm:-

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>
#define N 5
sem_t forks[N];
pthread_t philosophers[N];
void* philosopher(void* num) {
    int id = *(int*)num;
    printf("Philosopher %d is thinking\n", id);
    sleep(1);
    sem_wait(&forks[id]);
    sem_wait(&forks[(id + 1) % N]);
    printf("Philosopher %d is eating\n", id);
    sleep(1);
    sem_post(&forks[id]);
    sem_post(&forks[(id + 1) % N]);
    printf("Philosopher %d finished eating\n", id);
    return NULL;
}
int main() {
    int i, id[N];

    for (i = 0; i < N; i++)
        sem_init(&forks[i], 0, 1);
    for (i = 0; i < N; i++) {
        id[i] = i;
        pthread_create(&philosophers[i], NULL, philosopher, &id[i]);
    }
    for (i = 0; i < N; i++)
        pthread_join(philosophers[i], NULL);
    return 0;
}
```

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exp4.c

2026-02-10-file-1.term

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~\$ gcc exp4.c -o exp4

~\$./exp4

Philosopher 0 is thinking

Philosopher 1 is thinking

Philosopher 2 is thinking

Philosopher 3 is thinking

Philosopher 4 is thinking

Philosopher 0 is eating

Philosopher 2 is eating

Philosopher 4 is eating

Philosopher 0 finished eating

Philosopher 2 finished eating

Philosopher 1 is eating

Philosopher 4 finished eating

Philosopher 1 finished eating

Philosopher 3 is eating

Philosopher 3 finished eating

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